

Rati DEVIDZE

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OBJECTIVE: I am a Reinforcement Learning researcher and engineer with over 8 years of experience across academia and industry. I hold a Ph.D. focusing on Reward Design for Reinforcement Learning Agents from the Max-Planck Institute. My interests lie in the areas of Reinforcement Learning, Machine Learning, Data Science, AI, and Robotics.

EDUCATION

APRIL 2024	Ph.D. - Computer Science - Max Planck Institute for Software Systems (MPI-SWS) – Germany
NOV 2018	• Thesis: Reward Design for Reinforcement Learning Agents.
AUG 2018	Graduate School of Computer Science - Saarland University (UdS) – Germany
SEP 2016	• Completed Courses: <i>The Elements of Statistical Learning, Deep Learning, Information Retrieval and Data Mining.</i>
JULY 2016	B.Sc. - Computer Science - Tbilisi State University (TSU) – Georgia
SEP 2012	• GPA: 4/4 (Excellent) • Selected courses: <i>Algorithms and Data Structures, Statistics, Discrete Mathematics, Operation Research, Multivariable Calculus, Linear Algebra.</i> • Honors: President's Scholarship holder.

WORK EXPERIENCE

PRESENT	Reinforcement Learning & Machine Learning Engineer - minds.ai – Remote (Germany)
MAY 2024	• Enhanced the RL agent's predictive performance by reducing the dimensionality of the environment's state space. • Implemented Direct Preference Learning (DPL) to train the policy directly—without relying on traditional reinforcement learning algorithms. This approach significantly accelerated training and drove measurable improvements in KPIs within a highly complex semiconductor fabrication (FAB) environment.
MAY 2024	Ph.D. Research Assistant - Max Planck Institute (MPI-SWS) – Germany
NOV 2018	• I started a Reinforcement Learning research at the MPI-SWS. I conduct research focused on reward design in reinforcement learning [3, 1, 2]. • (Inverse-)Reinforcement learning, scientific publications, papers writing, leading research, built systems to gather, process, and analyze different reinforcement learning algorithms and data.
MAY 2018	Research Immersion Lab - Max Planck Institute (MPI-SWS) – Germany
FEB 2018	• I worked in Machine Teaching Group of Dr. Adish Singla. I contributed to our project with Dr. Parameswaran Kamalaruban where we worked on the iterative machine teaching problem in the context of the Markov Decision Process [4]. • I implemented a Markov Decision Process solver and performed experiments to evaluate theoretical results.
SEP 2017	Research Immersion Lab - Helmholtz Center for Information Security (CISPA) – Germany
JULY 2017	• Worked on accountability of security protocols. Analyzed the security of these protocols using the theorem prover Tamarin.
AUG 2016	.Net Software Developer - Prodware Group – Georgia
MAY 2016	• Implemented new software products for business ideas. • Added new features to existing software.
MAY 2016	Engineer Programmer - Akhali Kselebi Ltd – Georgia
MAR 2014	• Implemented desktop applications for the station operators. • Analysed international call statistics using ASP.NET charts • Implemented Windows and Web-services of the billing system for telecommunications

SKILLS

- **Reinforcement Learning:** Reward design; Preference Learning; Meta Reinforcement Learning; Inverse Reinforcement Learning; Large Language Models.
- **Machine Learning:** Deep Learning; Inferential Statistics; Applied, Algorithmic, and Theoretical Machine Learning; Optimization; Statistics; Data Science.
- **Soft Skills:** Scientific and Non-scientific Presentations; Teaching; Leadership; Self-management.
- **Languages:** Georgian (native); English (professional); German (C1); Russian (A2).

PROJECTS

- **Explicable Reward Design for Reinforcement Learning Agents:** Scientific work where we seek to capture two properties of reward functions: (a) informativeness so that the rewards speed up the agent’s convergence, and (b) sparseness so that the rewards are easy to interpret and debug [3, 2].
- **Exploration-Guided Reward Shaping for Reinforcement Learning under Sparse Rewards:** Scientific work where we used meta-gradient reinforcement learning methods to derive a framework that operates in a fully self-supervised manner and can accelerate an agent’s learning even in sparse-reward environments [1].
- **Curriculum Design for Teaching via Demonstrations:** Scientific work where we studied teaching via demonstrations in sequential decision-making settings. In particular, how to design a personalized curriculum over demonstrations to speed up the learner’s convergence [4, 5].

TEACHING

- **2021:** Teaching Assistant for Reinforcement Learning course at MPI-SWS and Saarland University.
- **2020:** Teaching Assistant for seminar course Multi-agent Reinforcement Learning at MPI-SWS and Saarland University.
- **2019:** Teaching Assistant for seminar course Machine Teaching at MPI-SWS and Saarland University.

HONORS AND AWARDS

- Scholarship at Graduate School of Computer Science at Saarland University (3% of the applicants are selected).
- Granted a German Academic Exchange Service (DAAD) scholarship in 2014 for German language training at RWTH Aachen University.
- President’s Scholarship holder at Tbilisi State University

SELECTED PUBLICATIONS

My work has been published at top venues in machine learning, including ICML, NeurIPS, IJCAI, AAMAS, and more. For a complete list of publications, kindly check my Google Scholar or DBLP profile.

- [1] Rati Devidze, Parameswaran Kamalaruban, and Adish Singla. Exploration-guided reward shaping for reinforcement learning under sparse rewards. In *Advances in Neural Information Processing Systems 36th Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022*.
- [2] Rati Devidze, Parameswaran Kamalaruban, and Adish Singla. Informativeness of reward functions in reinforcement learning. In *Proceedings of the 2024 International Conference on Autonomous Agents and Multiagent Systems, AAMAS 2024*.
- [3] Rati Devidze, Goran Radanovic, Parameswaran Kamalaruban, and Adish Singla. Explicable reward design for reinforcement learning agents. In *Advances in Neural Information Processing Systems 35th Annual Conference on Neural Information Processing Systems 2021, NeurIPS 2021*.
- [4] Parameswaran Kamalaruban, Rati Devidze, Volkan Cevher, and Adish Singla. Interactive teaching algorithms for inverse reinforcement learning. In *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence, IJCAI 2019, Macao, China, August 10-16, 2019*, pages 2692–2700, 2019.
- [5] Gaurav Yengera, Rati Devidze, Parameswaran Kamalaruban, and Adish Singla. Curriculum design for teaching via demonstrations: Theory and applications. In *Advances in Neural Information Processing Systems 35th Annual Conference on Neural Information Processing Systems 2021, NeurIPS 2021*.